

AcademiQ Student Data Attributes Specification

Institutional Data Dictionary, Schema Attributes, and Validation Rules for Accreditation & Analytics

RELEASE 2.4

NBA TIER-II COMPLIANT

Technical Standard Specification

Specification Scope: This document defines every required, computed, and accreditation-linked attribute for student master records in AcademiQ. It serves as the formal schema contract for batch Excel/CSV uploads (Student_Master_Data.xlsx), API data integrations, and database tables supporting Continuous Internal Evaluation (CIE), Attendance, ML Retention Risk Modeling, DPDP parent outreach, and NBA Criteria 4 & 5 reporting.

1. Primary Identity & Demographic Attributes

Table: students • Primary Master Entity

Core institutional attributes that uniquely identify every student and establish foundational academic records.

ATTRIBUTE KEY	DISPLAY NAME	TYPE / FORMAT	REQUIREMENT	VALIDATION RULE & SYSTEM USAGE	EXAMPLE VALUE
student_id / usn	University Seat No. (USN)	VARCHAR(50)	Mandatory	Primary unique identifier. Must match university roll format (e.g., 1RV22CS001). Unique index.	1RV22CS045
name	Student Full Name	VARCHAR(200)	Mandatory	Full legal name as registered in university admissions roster. Cannot be empty.	Aarav Sharma
email	Institutional Email	VARCHAR(255)	Mandatory	Valid email format. Used for portal sign-in, academic alerts, and timetable notices.	aarav@academiq.edu
phone	Mobile Number	VARCHAR(20)	Mandatory	10-digit mobile number with optional country code. Used for SMS & identity verification.	+91 9876543210
department_id	Department Code	VARCHAR(20)	Mandatory	Foreign key referencing academic department (e.g. CSE, ISE, ECE).	CSE
admission_year	Year of Admission	VARCHAR(10)	NBA Req	Academic entry cohort (e.g. 2022-23). Essential for NBA SAR Table 4.1 intake analysis.	2022-23
admission_quota	Admission Quota / Category	VARCHAR(50)	NBA Req	Category: KCET (Govt), COMED-K, Management, SNQ, Lateral Entry.	KCET
gender	Gender	VARCHAR(10)	Optional	Male, Female, Other. Used for institutional diversity reporting.	Male
mentor_faculty_id	Proctor / Mentor Faculty ID	VARCHAR(50)	NBA Req	Assigned faculty mentor ID (matches faculty.faculty_id). Required for Criterion 5.3.	FAC001

2. Current Enrollment & Section Allocation Attributes

Tables: students, student_enrollments

Locates student in current academic term, class division, and physical timetable room matrix.

ATTRIBUTE KEY	DISPLAY NAME	TYPE / FORMAT	REQUIREMENT	VALIDATION RULE & SYSTEM USAGE	EXAMPLE VALUE
semester	Current Semester	INTEGER (1 – 8)	Mandatory	Active semester of study (1 to 8). Governs curriculum courses, syllabus, and proctoring.	3
section	Section / Division	VARCHAR(10)	Mandatory	Assigned division: A, B, C. Maps student to daily timetable slots & classrooms.	A
academic_term	Academic Term	VARCHAR(20)	Mandatory	Current institutional session (e.g. 2024-25-EVEN or 2024-25-ODD).	2024-25-EVEN
batch_year	Graduating Batch	VARCHAR(15)	NBA Req	4-year cohort label (e.g. 2022–2026). Required for NBA cohort tracking across years.	2022-26
enrollment_status	Enrollment Status	VARCHAR(20)	Mandatory	Status: active, detained, discontinued, graduated.	active

3. Continuous Internal Evaluation (CIE) Course-Level Attributes

Field: students.courses_data (JSON Matrix)

Granular continuous assessment components adhering to autonomous university guidelines (CIE Raw: 100 → Reduced: 50).

ATTRIBUTE KEY	DISPLAY NAME	TYPE / BOUNDS	REQUIREMENT	WEIGHTAGE & VALIDATION FORMULA	EXAMPLE VALUE
course_code	Subject Code	VARCHAR(20)	Mandatory	Official university subject code (e.g. CS3C01, CS5C02, CS8E501).	CS3C01
course_name	Subject Title	VARCHAR(150)	Mandatory	Full course title as registered in academic legend.	Data Structures
credits	Course Credits	FLOAT (1.0 – 5.0)	Mandatory	Credit weightage: 4.0 (Core Theory), 3.0 (Elective), 1.5 (Laboratory). Used for SGPA.	4.0
cie1	Internal Test 1 (IA-1)	FLOAT (0 – 25.0)	Mandatory	Written assessment conducted at 1/3rd semester. Raw score out of 25. 'A' if absent.	21.5 / 25.0
cie2	Internal Test 2 (IA-2)	FLOAT (0 – 25.0)	Mandatory	Written assessment conducted at 2/3rd semester. Raw score out of 25. 'A' if absent.	23.0 / 25.0
cie3	Test 3 / Improvement	FLOAT (0 – 25.0)	Optional	Optional improvement test. Best 2 of 3 considered if institutional policy permits.	24.0 / 25.0
quiz1	Quiz Assessment 1	FLOAT (0 – 10.0)	Mandatory	Continuous objective/MCQ quiz conducted during regular classes. Raw score out of 10.	8.5 / 10.0
quiz2	Quiz Assessment 2	FLOAT (0 – 10.0)	Mandatory	Second continuous assessment quiz. Raw score out of 10.	9.0 / 10.0
el / asn	Experiential Learning	FLOAT (0 – 30.0)	Mandatory	Case study, mini-project, lab simulation, or assignment evaluation out of 30.	27.0 / 30.0
cie_raw	Total CIE Raw Sum	FLOAT (0 – 100.0)	Computed	Formula: CIE1 + CIE2 + Quiz1 + Quiz2 + EL (Sum out of 100).	89.0 / 100.0
cie_reduced	Reduced CIE Component	FLOAT (0 – 50.0)	Computed	Formula: cie_raw × 50 / 100. Internal assessment component out of 50.	44.5 / 50.0

4. Semester End Examination (SEE) & Grade Attainment

Standard 10-Point Absolute Scale

University letter grade mapping and grade point determination applied once final semester examinations conclude.

ATTRIBUTE KEY	DISPLAY NAME	TYPE / BOUNDS	REQUIREMENT	CALCULATION FORMULA & RULES	EXAMPLE VALUE
see_raw	Semester Exam Score (SEE)	FLOAT (0 – 100.0)	Mandatory	Final written semester examination score out of 100. Minimum 35 required to pass SEE.	78.0 / 100.0
see_reduced	Reduced SEE Component	FLOAT (0 – 50.0)	Computed	Formula: see_raw × 50 / 100. External exam component out of 50.	39.0 / 50.0
grand_total	Grand Course Total	FLOAT (0 – 100.0)	Computed	Formula: cie_reduced + see_reduced. Combined score out of 100. Minimum 50 to pass.	83.5 / 100.0
grade	Letter Grade	VARCHAR(2)	Computed	Scale: S (≥90), A (≥80), B (≥70), C (≥60), D (≥55), E (≥50), F (<50).	A
grade_points (gp)	Grade Points	INTEGER (0 – 10)	Computed	Points: S:10, A:9, B:8, C:7, D:6, E:5, F:0. Used for SGPA weighting.	9
course_status	Completion Status	VARCHAR(15)	Computed	completed (Passed), failed (Backlog), in_progress (During term).	completed

5. Cumulative Academic Progression & Risk Analytics

NBA Criterion 4.3 / 4.4 & ML Retention

Synthesized metrics evaluated across semesters to feed the Random Forest / XGBoost early warning model and generate NBA Tables 4.3 & 4.4.

ATTRIBUTE KEY	DISPLAY NAME	TYPE / BOUNDS	REQUIREMENT	TECHNICAL DEFINITION & PURPOSE	EXAMPLE VALUE
current_sgpa	Semester GPA (SGPA)	FLOAT (0.0 – 10.0)	Computed	Formula: $\Sigma(\text{Grade_Points} \times \text{Credits}) / \Sigma(\text{Credits})$ for all completed term courses.	8.75
cumulative_cgpa	Cumulative CGPA	FLOAT (0.0 – 10.0)	NBA Req	Overall cumulative GPA computed across all completed terms. Crucial for Criteria 4.4 & 4.5.	8.42
previous_gpa	Preceding Term GPA	FLOAT (0.0 – 10.0)	Mandatory	Immediate preceding semester SGPA. Primary predictive feature for trajectory analytics.	8.10
backlogs	Active Backlogs Count	INTEGER (≥ 0)	NBA Req	Count of currently uncleared courses (F grade). Determines Table 4.3 vs 4.4 classification.	0 (Clear)
progression_tier	Progression Status	VARCHAR(30)	NBA Req	Passed without Backlog, Passed with Backlog, Detained / Year-Back.	Clear Progression
risk_score	ML Risk Probability	FLOAT (0.0 – 1.0)	Computed	Predicted probability of academic failure output by XGBoost/Random Forest model.	0.14 (Low)
engagement	Engagement Level	VARCHAR(20)	Computed	Derived category: High, Medium, Low. Synthesized from LMS & class activity.	High

6. Course-wise Attendance & Duty Leave Accounting

Tables: students, class_attendance_entries

Classroom attendance telemetry. Directly governs exam hall ticket issuance (75% threshold) and integrates event duty leave credits.

ATTRIBUTE KEY	DISPLAY NAME	TYPE / BOUNDS	REQUIREMENT	VALIDATION RULE & INSTITUTIONAL POLICY	EXAMPLE VALUE
classes_conducted	Classes Engaged	INTEGER (≥ 0)	Mandatory	Total number of lecture / lab periods conducted by faculty for the course to date.	48
classes_attended	Classes Attended	INTEGER (≥ 0)	Mandatory	Total periods physically attended + officially approved duty leaves credited.	42
attendance_pct	Course Attendance %	FLOAT (0 – 100.0%)	Computed	Formula: (classes_attended / classes_conducted) × 100.	87.5%
semester_attendance_pct	Overall Term Attendance %	FLOAT (0 – 100.0%)	Computed	Arithmetic mean of attendance percentages across all enrolled semester courses.	89.2%
eligibility_status	Exam Eligibility Status	VARCHAR(25)	Computed	Eligible (≥75%), Condonation (65%–74.9%), Detained / NSAR (<65%).	Eligible
duty_leave_credits	Duty Leave Credited Hours	INTEGER (≥ 0)	Optional	Hours credited back from verified co-curricular hackathon / symposium attendance.	4 Hours

7. Parent / Guardian Contact & Privacy Attributes

Table: parent_contacts • DPDP Act 2023 Compliant

Maintained under strict privacy compliance to enable automated low-attendance warnings and proctor notifications via Twilio proxy bridging.

ATTRIBUTE KEY	DISPLAY NAME	TYPE / BOUNDS	REQUIREMENT	PRIVACY IMPLEMENTATION & PURPOSE	EXAMPLE VALUE
parent_name	Parent / Guardian Name	VARCHAR(200)	Mandatory	Full legal name of primary guardian authorized for institutional communications.	Rajesh Sharma
relationship	Relationship	VARCHAR(50)	Mandatory	Father, Mother, Local Guardian. Determines primary contact hierarchy.	Father
parent_phone	Primary Contact Number	VARCHAR(20)	Mandatory	Valid phone number. Never exposed in plaintext to end users; bridged via Twilio proxy.	+91 9845012345
parent_email	Parent Email Address	VARCHAR(255)	Optional	Electronic mail address for official term grade cards, fee dues, and proctor letters.	r.sharma@domain.com
preferred_language	Preferred Language	VARCHAR(30)	Optional	Language for automated IVR voice alerts: English, Kannada, Hindi.	English
alert_opt_in	Automated Alert Consent	BOOLEAN	Mandatory	Explicit consent record under DPDP Act 2023 for automated SMS and voice calls.	TRUE (Active)

8. Placement, Higher Studies & Student Achievements

Tables: placement_records, student_achievements

Tracks graduating student outcomes, competitive achievements, and club leadership required for institutional accreditation audits.

ATTRIBUTE KEY	DISPLAY NAME	TYPE / BOUNDS	REQUIREMENT	ACCREDITATION CRITERION & PURPOSE	EXAMPLE VALUE
placement_status	Placement Outcome	VARCHAR(50)	NBA Req	Placed, Higher Studies, Entrepreneurship, Seeking. (Criterion 4.5).	Placed
company_name	Recruiting Organization	VARCHAR(200)	NBA Req	Name of employer issuing offer letter. Mapped to campus placement statistics.	Oracle Financial
package_etc	Annual CTC (INR LPA)	FLOAT (LPA)	NBA Req	Remuneration in Lakhs Per Annum. Computes median and average salary for NIRF & NBA.	14.5 LPA
offer_letter_ref	Offer Letter Document Proof	VARCHAR(500)	NBA Req	Document storage reference for audit verification by NBA peer visiting committee.	doc_offer_STU001.pdf
higher_studies_details	Higher Education / Exam	VARCHAR(250)	NBA Req	Admitted program (e.g. M.S. CS) + Qualifying exam score (GATE, GRE, CAT).	GATE Score: 720 (IIT-M)
club_roles	Student Club Leadership	VARCHAR(50)	NBA Req	Leadership role: head, council, member. (Criterion 4.6 Professional Activities).	Head, ACM Chapter
awards_honors	Student Achievements	TEXT	NBA Req	Hackathon awards, research publications, athletic achievements with proof certificates.	1st Prize, Smart India Hackathon

9. Batch Ingestion Specifications & Excel Standards

Supported Formats: .xlsx, .csv, .json

When preparing spreadsheets (e.g., Student_Master_Data.xlsx) for automatic ingestion into AcademiQ, the following rules apply:

Special Symbol Notation in Grade Reports

- **'A':** Indicates student was *Absent* for that test (computed as 0.0 with absent flag recorded).
- **'NE':** Indicates *Not Eligible / Not Entered* (student barred due to attendance deficit).
- **'NA' (Shaded Cells):** Student was not enrolled for that elective course or semester report.
- **USN Header:** Column header must be labeled USN or student_id in row 1 or 2.

Database Normalization & Multi-Report Matching

- **Primary Anchor:** All disparate reports (Attendance, IA marks, Electives) join cleanly on USN.
- **Header Regex:** Course code in header must match pattern (CS[0-9][A-Z][0-9]{2}).
- **Section Normalization:** Automatic mapping to sections A, B, C based on division tags.
- **Idempotent Upsert:** Re-uploading spreadsheets updates existing marks without duplicate rows.

10. NBA SAR Criteria Traceability Matrix

Tier-II Engineering Criteria Linkage

Maps every student data attribute directly to its corresponding NBA Self-Assessment Report (SAR) table and formula.

SAR CRITERION	CRITERION DESCRIPTION	REQUIRED STUDENT DATA ATTRIBUTES	ACCREDITATION METRIC
Criterion 4.1	Enrolment Ratio	admission_year, admission_quota, department_id	Intake vs. Admitted % (N1/N)
Criterion 4.2	Success Rate Without Backlogs	admission_year, backlogs, grade, batch_year	% Passing in 4 years without backlogs
Criterion 4.3	Success Rate With Backlogs	admission_year, backlogs, cleared_backlogs, cgpa	% Passing within stipulated + 1 year
Criterion 4.4	Academic Performance Index (API)	current_sgpa, cumulative_cgpa, see_raw, credits	Mean CGPA in 2nd, 3rd, and 4th Years
Criterion 4.5	Placement, Higher Studies & Entrepreneurship	placement_status, company_name, package_ctc, offer_letter_ref	Placement Index (X + Y + Z) / N
Criterion 4.6	Professional Activities & Student Events	club_roles, event_registrations, awards_honors	Student Chapters & Hackathon Attainment
Criterion 5.3	Faculty Proctoring & Mentoring	mentor_faculty_id, attendance_pct, parent_phone	Mentoring ratio (1:15–20) & intervention logs

11. Database Entity-Relationship Mapping Summary

PostgreSQL Relational Storage

Core Relational Entities

- students: Primary master table containing USN, demographics, attendance %, internal marks, and courses JSON.
- departments: 1-to-N parent entity for student program enrollment.
- student_enrollments: Term-specific mapping linking student to academic section and timetable rooms.
- parent_contacts: 1-to-1 privacy-protected guardian contact records.

Accreditation & Outcome Entities

- placement_records: Placement offers, packages, and company details.
- student_achievements: Certified student honors, competitions, and prizes.
- event_registrations: Co-curricular RSVP, attendance, and duty leaves.
- academic_batches / progress: Cohort progression for Criteria 4.1 to 4.4.